FORTIGATE NSE4

Lab 1 — Initializing FortiGate
Creating Users & Admin Profiles

A complete lab walkthrough covering FortiGate VM deployment on EVE-NG — from downloading the firmware image and deploying the topology, through initial CLI and GUI setup, interface configuration, system settings hardening, to creating custom administrator accounts and read-only profiles.

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Category: FortiGate NSE4 Lab Series · Portfolio Project
Platform: FortiGate VM v7.6.5 on EVE-NG · NAT/Routed Mode

1. Download FortiGate VM Image

The first step is obtaining the FortiGate VM image from FortiCloud for deployment on EVE-NG.

1. Log in to FortiCloud (https://support.fortinet.com) with your FortiCare credentials.

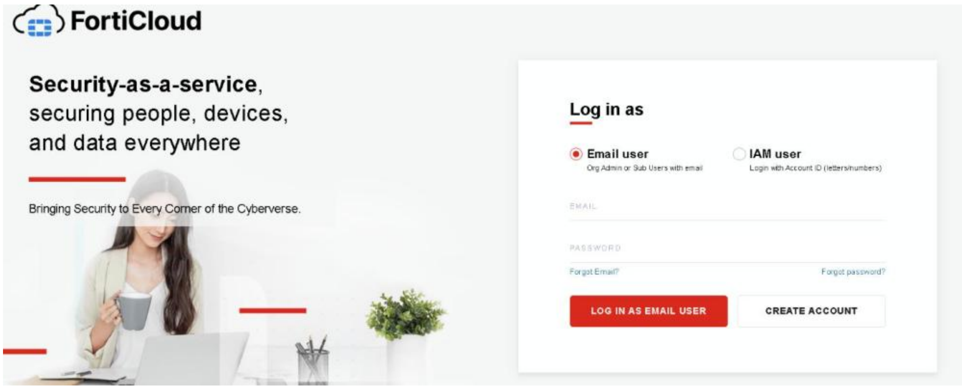


Figure 1 — FortiCloud login page

2. Navigate to Support → VM Images → select FortiGate as the product and KVM as the platform.

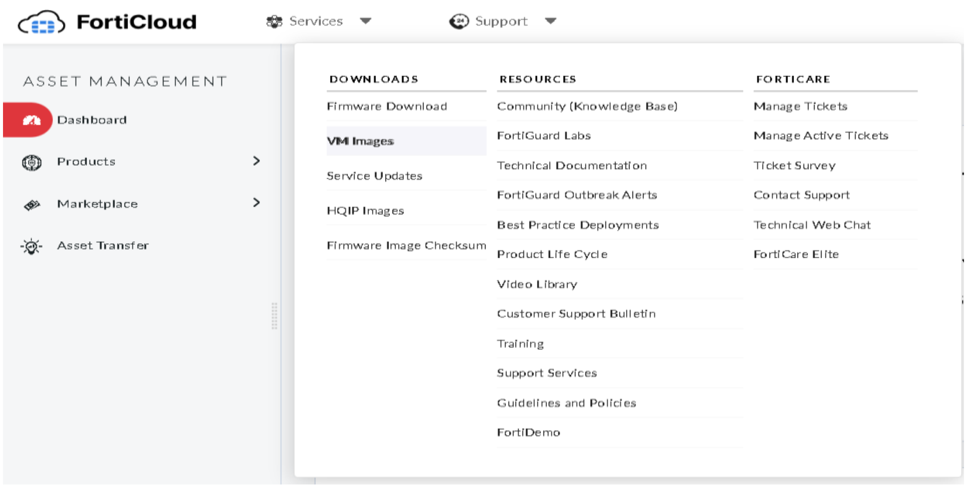


Figure 2 — FortiCloud: VM Images download page

3. Download the latest version (v7.6.5) — select the "New deployment" KVM image.

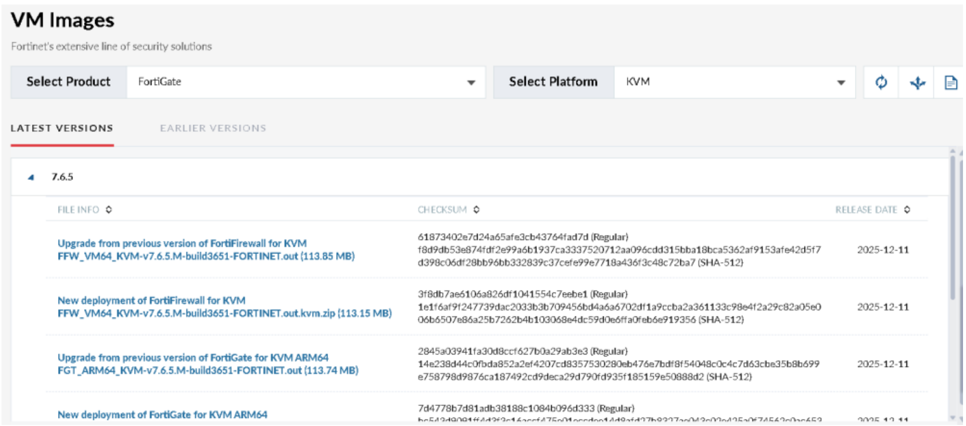


Figure 3 — FortiGate v7.6.5 KVM image with checksums

4. Rename the folder to fortinet-7.6.5 and convert the image to virtioa.qcow2.
5. Deploy the image to EVE-NG.

2. Lab Topology

The lab uses a three-port topology following best practice by separating management traffic from user traffic:

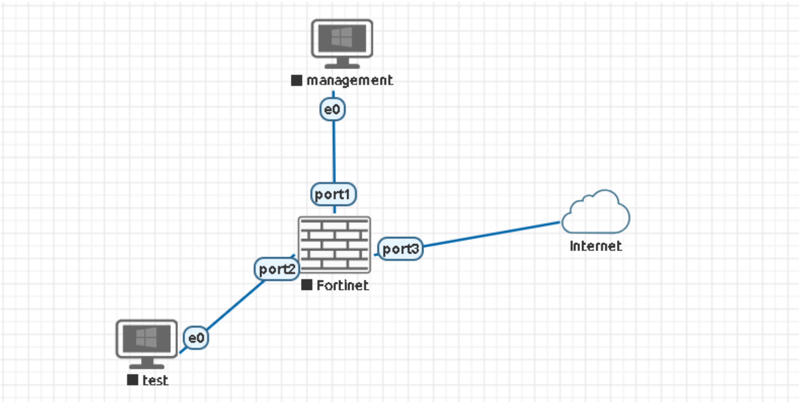


Figure 4 — EVE-NG Lab Topology

Port	Role	Connected To
port1	MGMT (Management)	Management PC
port2	LAN (Inside)	Test PC
port3	WAN (Outside)	Internet (Cloud)

This separation is best practice — management traffic (SSH, HTTPS admin) should never share the same interface as user data traffic, especially in OT environments.

3. Initial Login & Interface Configuration

3.1 First Login

Access the FortiGate CLI through the EVE-NG console. The default credentials are:

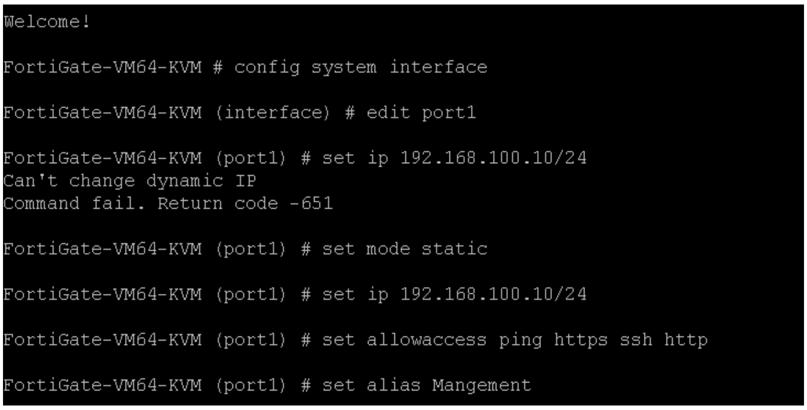
- **Username:** admin
- **Password:** (empty — press Enter)

You will be prompted to set a new password immediately. After setting the password, log in again with the new credentials.

3.2 Configure Management Interface (port1)

Port1 is dedicated to management access only. Configure it with a static IP, enable administrative access protocols, and set an alias:

```
config system interface
  edit port1
    set mode static
    set ip 192.168.100.10/24
    set allowaccess ping https ssh http
    set alias MANAGEMENT
  end
```



```
Welcome!

FortiGate-VM64-KVM # config system interface

FortiGate-VM64-KVM (interface) # edit port1

FortiGate-VM64-KVM (port1) # set ip 192.168.100.10/24
Can't change dynamic IP
Command fail. Return code -651

FortiGate-VM64-KVM (port1) # set mode static

FortiGate-VM64-KVM (port1) # set ip 192.168.100.10/24

FortiGate-VM64-KVM (port1) # set allowaccess ping https ssh http

FortiGate-VM64-KVM (port1) # set alias Mangement
```

Figure 5 — CLI: port1 management interface configuration

Note: If you encounter "Can't change dynamic IP" when setting the IP, first run set mode static to switch from DHCP to static addressing.

3.3 Configure Management PC

Assign the management PC a static IP on the same subnet:

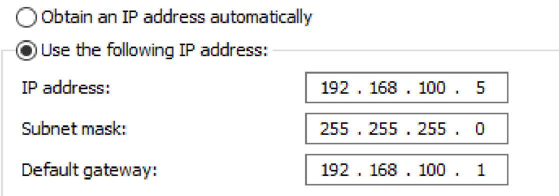
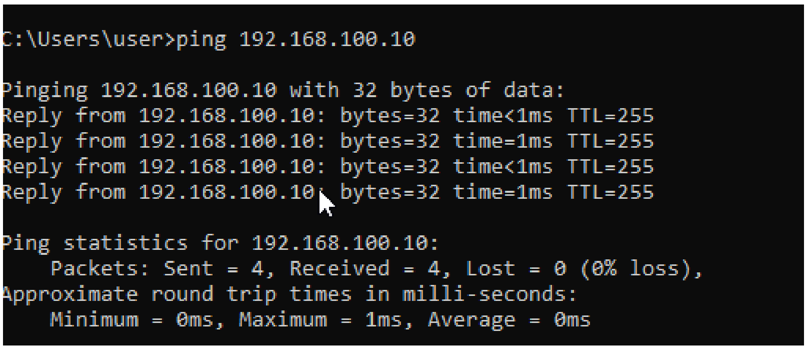


Figure 6 — Management PC: IP 192.168.100.5/24, gateway 192.168.100.1

3.4 Verify Connectivity

Ping the FortiGate management IP from the Management PC to confirm Layer 3 reachability:



```
C:\Users\user>ping 192.168.100.10

Pinging 192.168.100.10 with 32 bytes of data:
Reply from 192.168.100.10: bytes=32 time<1ms TTL=255
Reply from 192.168.100.10: bytes=32 time=1ms TTL=255
Reply from 192.168.100.10: bytes=32 time<1ms TTL=255
Reply from 192.168.100.10: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 7 — Successful ping to 192.168.100.10 (0% loss)

4. GUI Access & Initial Setup

4.1 Access the Web GUI

Open a browser on the Management PC and navigate to https://192.168.100.10/login. Log in with the same credentials used for the CLI:

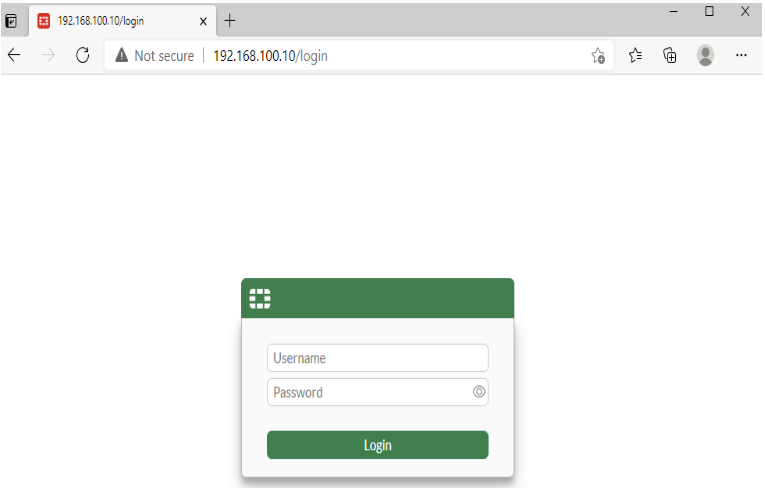


Figure 8 — FortiGate web GUI login page

4.2 License & FortiCare Registration

The GUI will prompt for licensing. Select the evaluation license option and enter your FortiCare credentials. The system will reboot after registration:

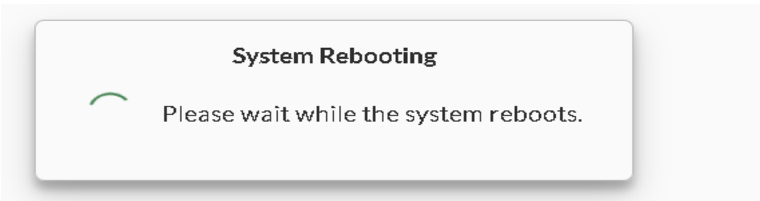


Figure 9 — System Rebooting after license registration

4.3 Disable Automatic Patching

For OT environments, automatic patching should be disabled. Patches must be tested and applied manually during maintenance windows to avoid unexpected disruptions to industrial processes:

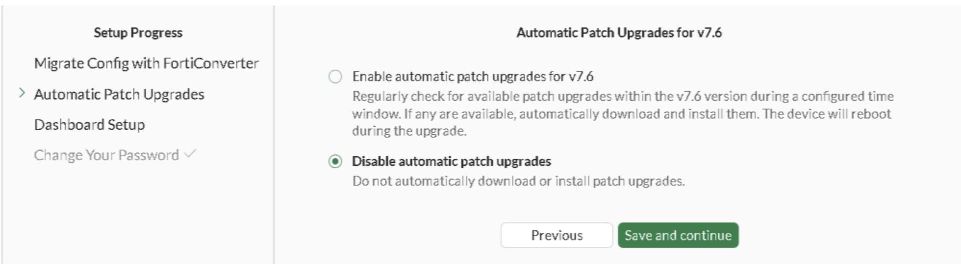


Figure 10 — Disable automatic patch upgrades (OT best practice)

4.4 Dashboard Overview

After the reboot, the FortiGate dashboard shows key system information. The appliance runs in NAT mode (routed mode, supporting Layer 3):

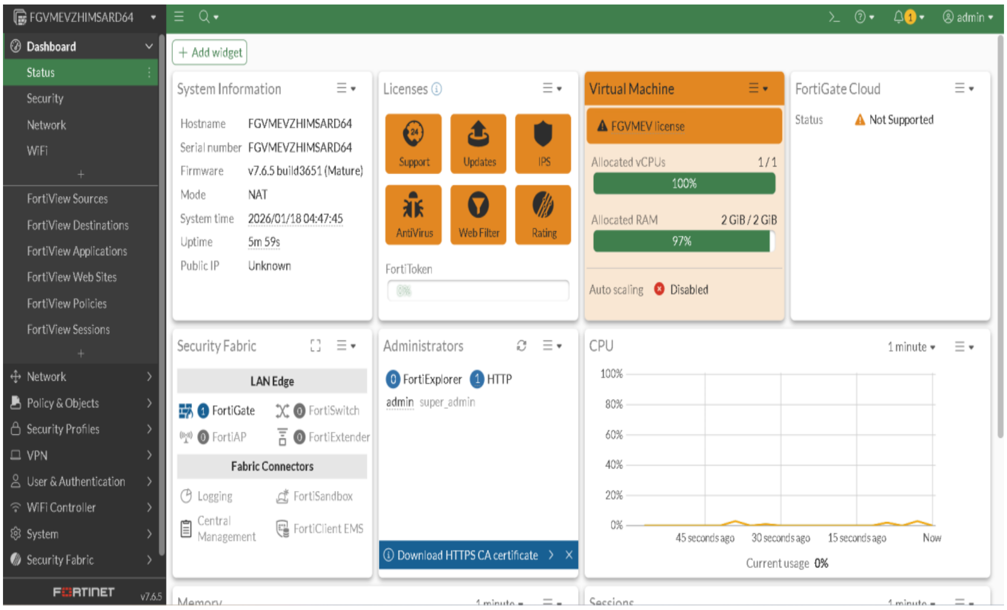


Figure 11 — FortiGate v7.6.5 dashboard (NAT mode)

Key dashboard widgets:

- System Information — hostname, serial, firmware version, mode, uptime
- Licenses — FortiGuard subscription status (evaluation in lab)
- Virtual Machine — vCPU and RAM allocation
- CPU / Memory — real-time usage graphs
- Sessions — active connection count
- Administrators — currently logged-in admin sessions

4.5 Dashboard Customization

You can add FortiView widgets (application bandwidth, source/destination analysis, policy views) and access the CLI Console directly from the dashboard:

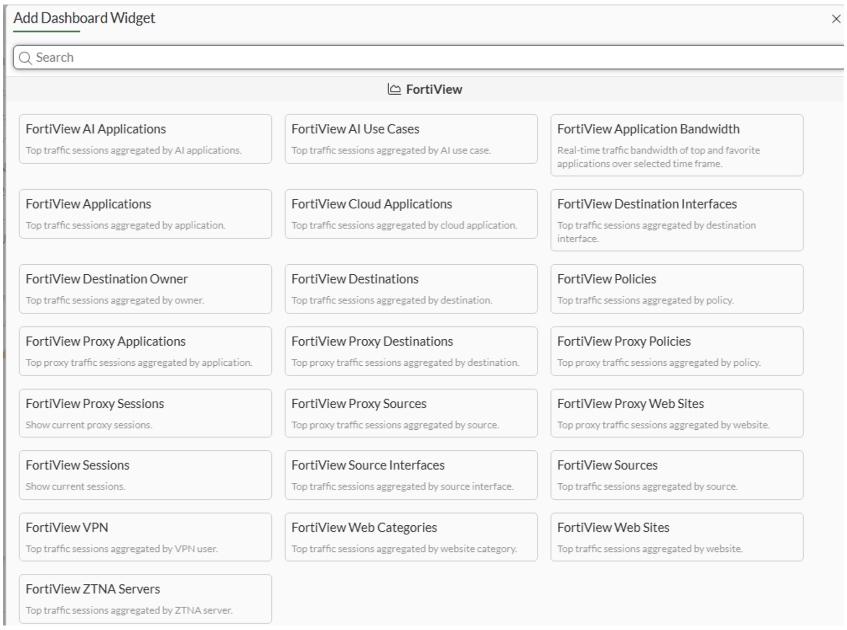


Figure 12 — Add Dashboard Widget: FortiView options

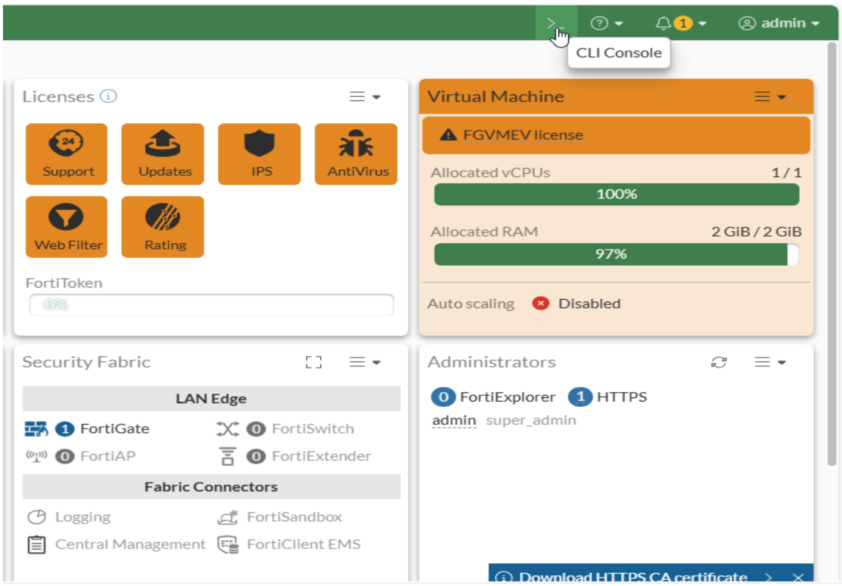


Figure 13 — CLI Console and License/VM status widgets

5. System Settings

Navigate to System → Settings to configure essential system parameters:

5.1 Hostname

Change the default hostname to something meaningful for your environment:

Figure 14 — System Settings: Hostname

5.2 Time Zone & NTP

Set the correct timezone and configure NTP synchronization. Accurate time is critical for log correlation and certificate validation:

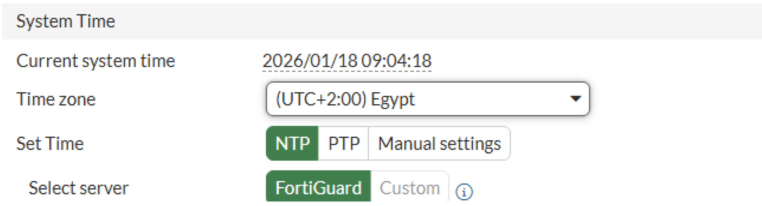


Figure 15 — System Time: timezone and NTP configuration

5.3 Administration Ports

The default HTTP (80) and HTTPS (443) admin ports can be changed, though this is generally not recommended unless required by security policy:

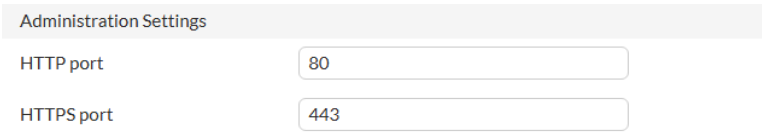


Figure 16 — Administration Settings: HTTP/HTTPS ports

5.4 Password Complexity Policy

Enforce strong password requirements for all administrator accounts:

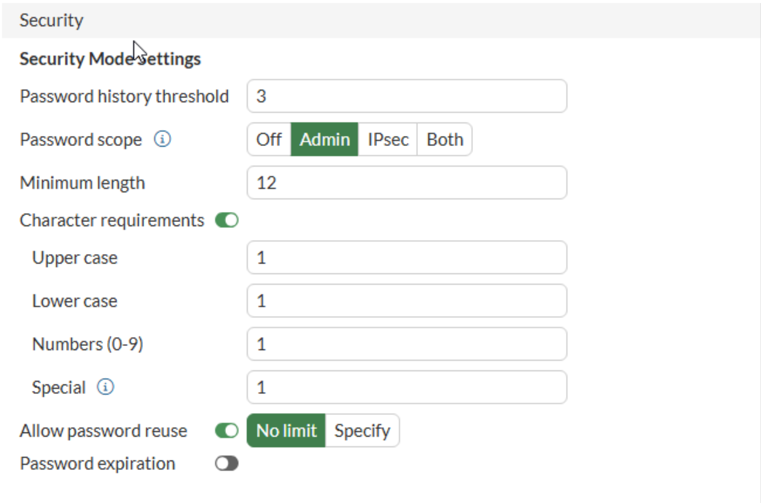


Figure 17 — Password policy: min 12 chars, upper/lower/number/special required

Setting	Value
Password history threshold	3 (cannot reuse last 3 passwords)
Password scope	Admin
Minimum length	12 characters
Upper case required	1
Lower case required	1
Numbers required	1
Special characters required	1
Allow password reuse	No limit
Password expiration	Disabled (or set per policy)

5.5 Workflow Management

Configuration save mode can be set to Automatic (changes saved immediately) or Manual (requires explicit save). For production OT environments, Manual mode adds a safety net against accidental changes:

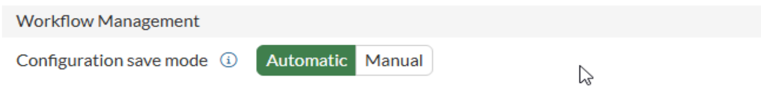


Figure 18 — Workflow Management: Automatic vs Manual save

5.6 Theme & View Settings

Customize the GUI language, theme, and date/time display format:

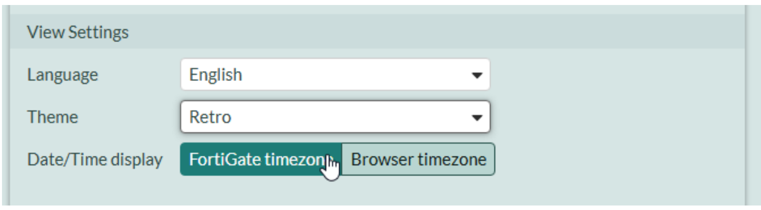


Figure 19 — View Settings: Language, Theme (Retro), Date/Time display

Remember to click APPLY after making any changes in System Settings.

6. Administrator Accounts & Profiles

6.1 Default Admin Account

FortiGate ships with a single "admin" account using the super_admin profile. This username is highly predictable and should be replaced in production:

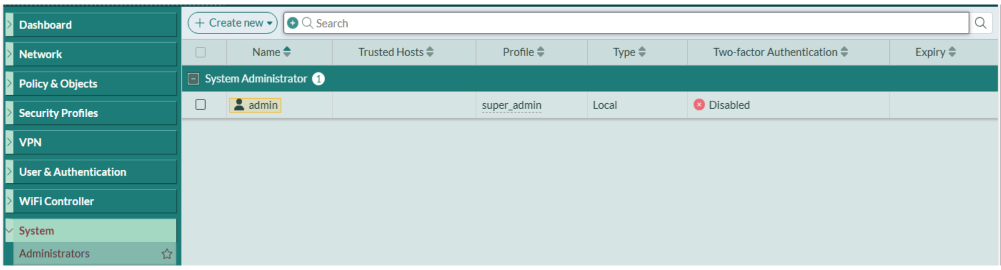


Figure 20 — System → Administrators: default admin account

6.2 Create a New Super Admin

Create a new administrator account with a non-guessable username but the same super_admin profile. FortiGate supports multiple authentication types:

- Local User — credentials stored on the FortiGate
- Match a user on a remote server group — RADIUS, LDAP, TACACS+
- Match all users in a remote server group
- Use public key infrastructure (PKI) group — certificate-based

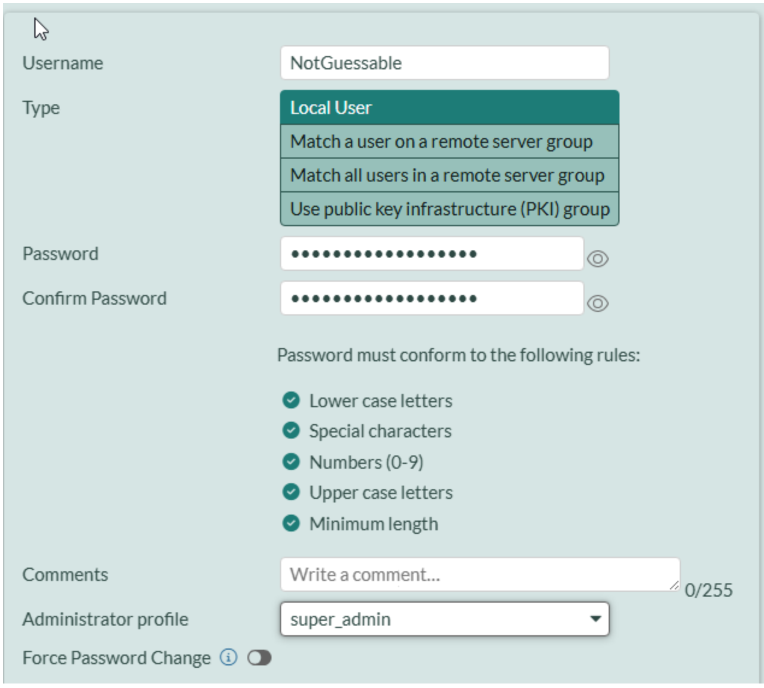


Figure 21 — New Administrator: NotGuessable (super_admin profile)

After creating the new super_admin account, log in with it and delete the default "admin" account. Note: you cannot delete or rename the account you are currently logged in as.

6.3 Create a Read-Only Admin Profile

For SOC analysts or auditors who need visibility but should not modify configurations, create a Read-Only profile:

1. Navigate to System → Admin Profiles.
2. Click Create New.
3. Name the profile (e.g., Read-Only).
4. Use Set All → Read to set all access control categories to Read permission.

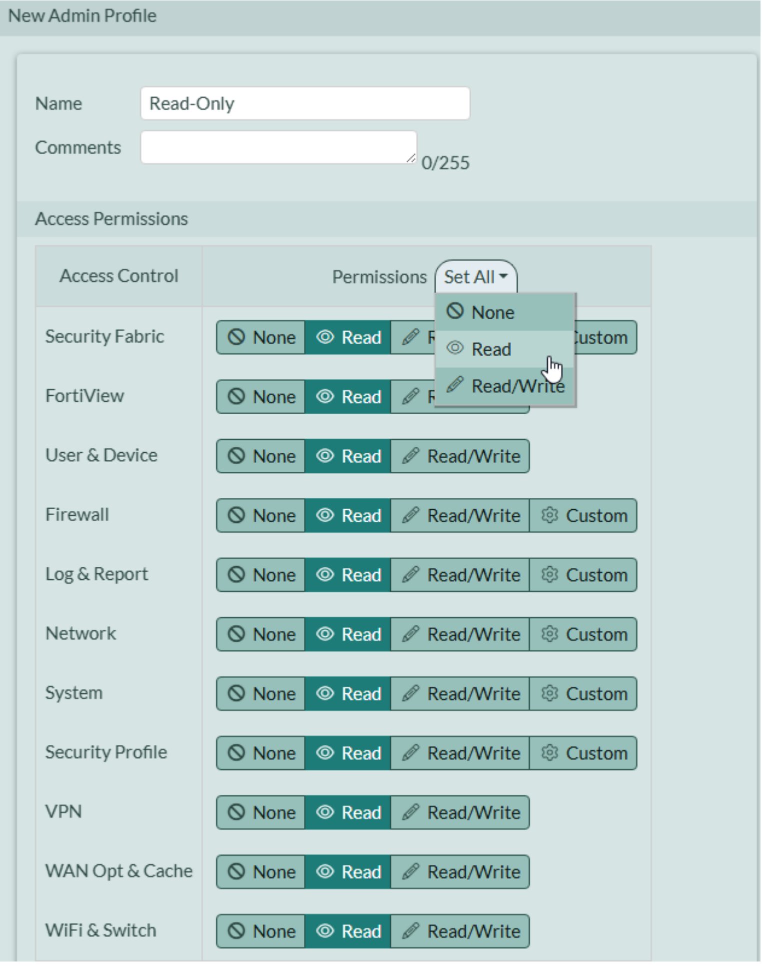


Figure 22 — New Admin Profile: Read-Only (all categories set to Read)

Available permission levels per category:

Permission	Description
None	No access to this feature category
Read	Can view but cannot modify
Read/Write	Full access to view and modify
Custom	Granular per-feature control (Firewall, Log, Network, System, Security Profile)

6.4 Create a Read-Only User Account

Create a new administrator and attach the Read-Only profile:

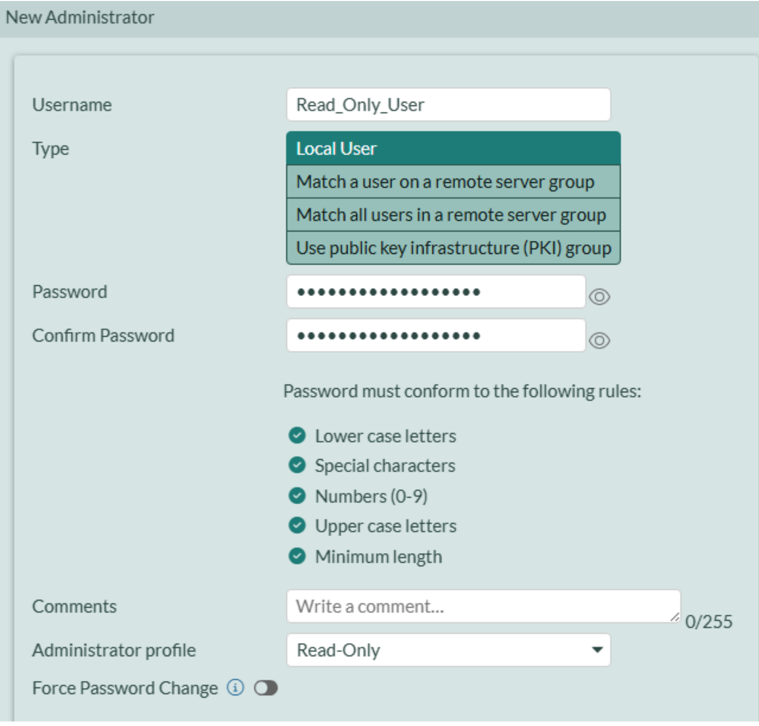


Figure 23 — New Administrator: Read_Only_User with Read-Only profile

7. Summary

Component	Details
FortiGate Version	v7.6.5 build3651 (Mature)
Deployment Platform	EVE-NG (KVM / virtioa.qcow2)
Operating Mode	NAT (Routed, Layer 3)
Management Interface	port1 — 192.168.100.10/24
LAN Interface	port2 (for user/test traffic)
WAN Interface	port3 (internet-facing)
Admin Access Protocols	ping, https, ssh, http on port1
Password Policy	12 chars min, upper/lower/number/special
Auto-Patching	Disabled (OT best practice)
Workflow Save Mode	Automatic or Manual
Admin Profiles	super_admin (full), Read-Only (custom), Custom
Best Practice	Replace default admin account with non-guessable username